

Introduction to competence descriptions

Digital competence in early childhood education and care and pre-primary education ICT competence in basic education

The purpose of these competence descriptions is to translate the valid National core curriculum for early childhood education and care and the National core curricula for basic education and pre-primary education into more concrete terms regarding digital competence. Education and ECEC providers will update their digital curricula to correspond with these competence descriptions.

As the digital world changes, so do competence needs. The first version of these descriptions was published in February 2021. Competence descriptions 2.0 were published in June 2022. The purpose of the modifications is to improve the quality of the descriptions and, in particular, make them easier to read and keep them up to date. Some deletions and additions have been made in the descriptions. An effort has been made to take into account users' diverse digital services and devices.

Digital competence and ICT competence as a whole consist of four main areas: 1) Practical skills and personal production, 2) Responsible and safe use of ICT, 3) Information management and inquiry-based and creative work, and 4) Interaction. The main areas were named consistently with the National core curriculum for basic education (2014). Following an update of the National core curriculum for ECEC (2022), the heading Digital competence has been introduced in the descriptions concerning early childhood education and care and pre-primary education.

The **digital services** referred to in the descriptions mean applications and software used in teaching and early childhood education and care. More extensively, **a digital environment refers** to all digital solutions, services, hardware and devices used in teaching and early childhood education and care. **A text** refers to texts that, according to the broad definition of text, may be information expressed with systems of verbal, visual, auditive, numeric and kinaesthetic symbols and their combinations. Where reference is made to a digital service and environment, the education and ECEC provider can in many cases identify more specifically the service and device they recommend that their staff and learners use.

We would like to encourage diverse use of digital environments! New technologies enable learners of different ages to assume the role of an active agent, producer and explorer.

The education and ECEC provider is responsible for ensuring that the services to be introduced are compliant with the EU's General Data Protection Regulation and national legislation. Teachers, contact the education and ECEC provider that employs you for more information!

For the competence descriptions, see the tabs (main areas 1 to 4).

The descriptions have been designed to form a continuum of cumulating competence.

Instructions for reading the descriptions:

The descriptions for early childhood education and care and pre-primary education should be read as a single entity.

The description for grades 1 to 2 can be read on its own or in parallel with the descriptions for early childhood education and care and pre-primary education.

The description for grades 3 to 6 should be read in parallel with the descriptions for grades 1 to 2.

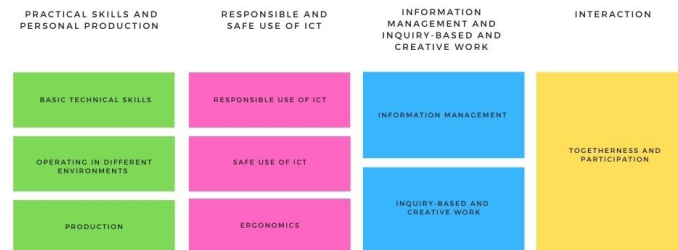
The description for grades 7 to 9 should be read in parallel with the descriptions for grades 1 to 2 and grades 3 to 6.

For more information about the descriptions of digital competence and ICT competence, please contact:

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To develop the descriptions further, we will be happy to receive feedback on them.

DIGITAL COMPETENCE



In pre-primary education

In early childhood education and care



In grades 7 to 9, the pupil

In grades 3 to 6, the pupil

In grades 1 to 2, the pupil



Operating in different environments

Digital competence

ICT competence

1B

In pre-primary education

In early childhood education and care

...the staff are familiar with and use digital services relevant to the child's early childhood education and care and pre-primary education appropriately.

...digital environments are used as part of support for development and learning.

...digital environments are actively used in pedagogical documentation.

...digital environments are used with a multi-sensory approach in the learning environment.

...children are familiarised with applications through exploration and play and through the themes of the learning areas.

...the staff play and examine the logic of games with the children. The staff modify familiar games and come up with new ones with the children.

...the staff use the gameful elements of digital environments.

...applications are used diversely and their main operating principles are observed.

In grades 7 to 9, the pupil

...uses digital environments naturally and diversely as part of studying and in ways specific to different subjects.

...is able to select a digital environment that is appropriate and suitable for him or her.

...uses digital services appropriately for studying.

...is able to transfer and process information between digital environments.

...is able to use digital services to receive and hand in assignments.

...is able to use society's digital services relevant to him or her.

...evaluates the progress of his or her learning with a feature available in a digital service.

...uses digital services fluently, on different devices if necessary.

...is familiar with career options related to different areas or digital competence.

...is familiar with the impacts of digitalisation on different fields.

...to understand the possibilities of the Internet of Things (IoT).

*...encourages other pupils to use digital environments diversely.

*...knows how to compare, evaluate and propose the use of digital environments.

*...is able to look for useful digital environments and explore the purposes for which they are used. The pupil knows how to register as a user of a service if necessary.

*...understands what it means when an application requests ri

*...knows how to download applications he or she finds interesting from the app store.

...is able to open and exit applications and move between them.

...with guidance, is able to move between the desktop and mobile version of the same service.

...practices perceiving the progress in his or her learning with the help of a feature in a digital service.

...with guidance, is able to use digital services to receive and hand in assignments.

...is familiar with the digital environments used in the school and knows how to sign into them. The pupil knows how to use digital services in a way that is sufficient for the grade he or she is in.

...is able to sign into any pupil management system that may be in use and familiarises himself or herself with its use.

In grades 3 to 6, the pupil

In grades 1 to 2, the pupil

...is able to use digital services that are important in his or her everyday life and understands their main operating principles.

...understands and is able to change rights in digital services.

...is able to use digital services (external to the teaching environment) that are used locally and contain information that is important from the point of view of the pupil's school work.

...with guidance, is able to use systems (external to the teaching environment) that are used locally and contain information that is important from the point of view of the pupil's school work.

...with guidance, is able to assign rights in digital services.

In pre-primary education

In early childhood education and care

- ...children's developing writing and reading skills are supported with digital environments.
- ...children film, record on video and modify content using simple functionalities.
- ...children play games that encourage them to produce their own content.
- ...digital environments are used in story crafting.
- ...children produce playful animations.
- ...children make personal and shared digital books underpinned by a broad definition of text.
- ...digital environments are used in playing with language.
- ...children practise playful reading and writing.
- ...complete photography and video projects.
- ...children draw up simple diagrams.
- ...children produce texts in accordance with the broad definition of text.
- ...children produce and interpret messages in digital environments.
- ...process photos and videos diversely through play and exploration.

In grades 7 to 9, the pupil

...käyttää digitaalisten ympäristöjen monipuolisia mahdollisuuksia tuottamisessa.

In grades 3 to 6, the pupil

In grades 1 to 2, the pupil

- ...uses image and video processing software diversely.
- ...knows how to store and share content in a cloud service. The pupil knows how to send files if necessary.
- ...takes photos, records videos and sound and carries out simple editing.
- ...with guidance, knows how to produce a work that contains text, sound, images and video image.
- ...uses a camera and other easy applications to make and record observations.
- ...practises commenting on content produced by others.
- ...knows how to edit content in a digital environment together with others.
- ...is able to comment and give feedback.
- *...is able to propose improvements that make presentations easier to understand using the possibilities provided by digital services.
- *...knows how to create a file for shared use and how to link other pupils to it.
- *...is familiar with and uses programs for editing images, video and sound. The pupil produces and shares edited content on different topics.
- *...knows how to keyword and use hashtags to organise content.
- ...knows how to arrange text, images and tables appropriately.
- ...knows how to take part in a joint production process of written or other content in a digital service.
- ...is able to design a logical folder and file structure. The pupil knows where to find the content he or she needs.
- ...is able to use styles in text. The pupil finds the special characters he or she needs.
- ...knows how use an automatic referencing tool to make citations and a list of references.
- ...is able to use cell references, formulas and the most common functions of a spreadsheet program.
- The pupil is able to draw up an appropriate diagram from statistical data by using a spreadsheet program.
- ...is able to draw up extensive, structured presentations using a presentation graphics program.
- ...masters the basics of presentation graphics. The pupil is able to draw up a presentation.
- ...masters the basics of spreadsheets.
- ...knows how to draw up a table and convert it into a diagram.
- ...knows how to edit text: activation of text, selection of font size and type.
- ...masters basic file processing skills: saving, copying, transferring and deleting.
- ...practises producing text in a digital service (multimodal texts).
- *...osaa käyttää kosketusnäyttökynää tarkoituksenmukaisesti.
- *...is familiar with different digital services intended for personal production and knows how to use them. The pupil is interested in editing images, videos and sound, and produces content independently.
- ...practises finding saved content.
- ...is able to work independently and together with others in the school's digital environments.
- ...knows how to produce text in digital environments. The pupil knows how to produce text for different purposes (multimodal texts).
- ...is able to find saved or shared content. The pupil understands about the folder and file structure when this is required by the operating system.
- ...masters the basics of a word processing application.

Responsible use of ICT

Digital competence

ICT competence

2A

In pre-primary education

In early childhood education and care



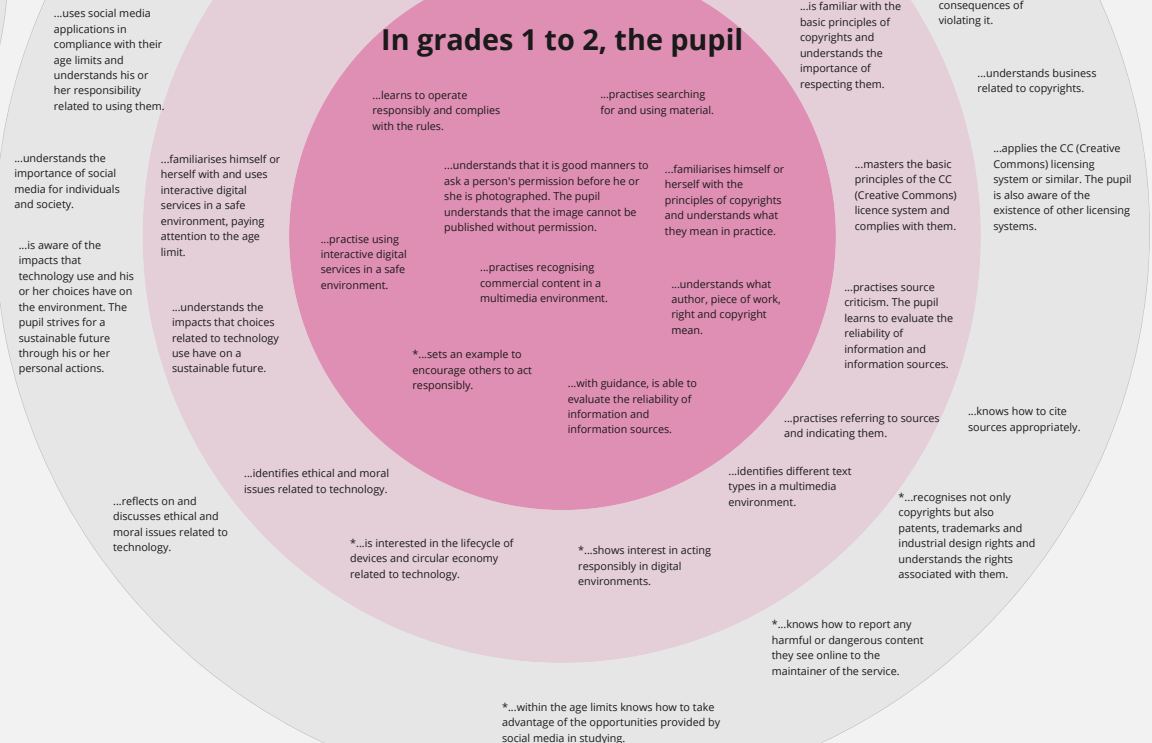
In grades 7 to 9, the pupil

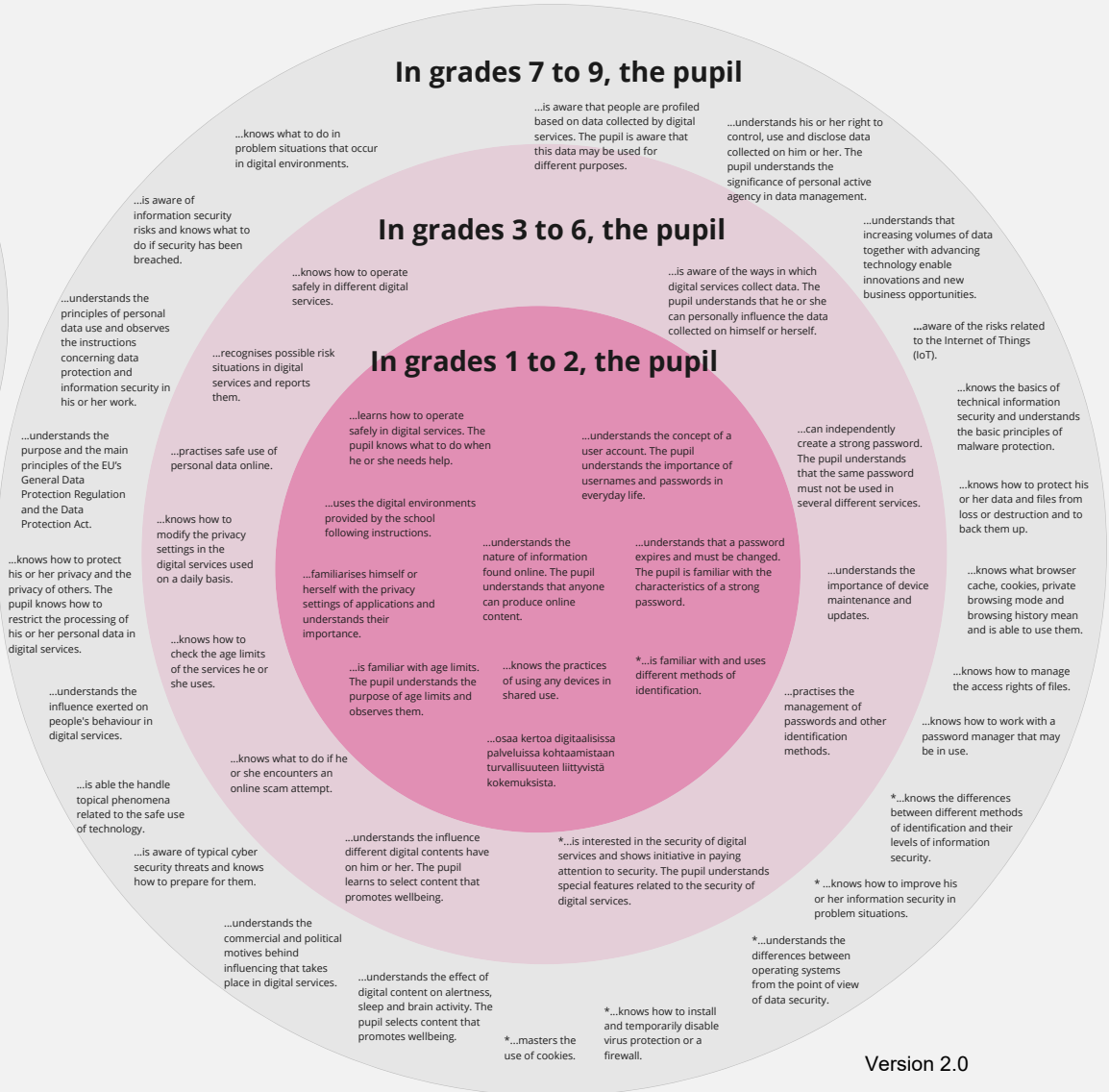
...understands the opportunities and risks related to technology in a global world.

In grades 3 to 6, the pupil

...knows what responsible operation means. The pupil acts responsibly and in compliance with the rules in digital services.

In grades 1 to 2, the pupil





In pre-primary education

In early childhood education and care

...the content offered to the children is of a high quality from the point of view of both user-centredness and pedagogy.

...children gather experiences of sustained work in digital environments.

...devices and digital environments that are suitable for children are used with the children.

...digital environments that encourage physical activity are favoured.

...the staff ensure that the hardware and devices are available to the children in a way that encourages them to use them in playing and learning.

...children learn to pay attention to lighting and the brightness of the display and to adjust them.

...the child's attention is drawn to taking breaks while working and to good working postures.

...children learn to pay attention to the quality and volume of sound.

...children stretch, exercise and take physical activity often and regularly.

...children practice ensuring good ergonomics and taking breaks.

In grades 7 to 9, the pupil

...is able to take into account the different aspects of ergonomics in his or her work.

...is able to focus his or her work in a goal-oriented and sustainable way in digital environments.

In grades 3 to 6, the pupil

...understands the importance of ergonomics for wellbeing in daily work.

...is able to choose a suitably long period of time for working.

In grades 1 to 2, the pupil

...practises healthy working postures.

...practises working in a goal-oriented manner in digital environments.

...knows how to adjust his or her workstation correctly to the extent that this is possible.

...practises taking breaks while working and having physical activity breaks.

...knows how to adjust the brightness, resolution and colour of the display and the volume.

...practises using digital environments that increase physical activity.

*...knows how to select the best tool and place for working from the point of view of ergonomics.

...finds out about suitable lighting.

*...is interested in technology that encourages physical activity.

...is able to use different ease of access and accessibility tools in digital environments.

*...knows how to find aids that improve ergonomics and understands their importance.

*...knows how to change the mouse cursor speed, the sensitivity of the touch screen and other control settings.

*...knows how to use the different input and control methods of devices. The pupil knows how to activate a suitable input and control method for the situation.

*...knows how to activate and disable ease of use and accessibility options.

In pre-primary education

In early childhood education and care

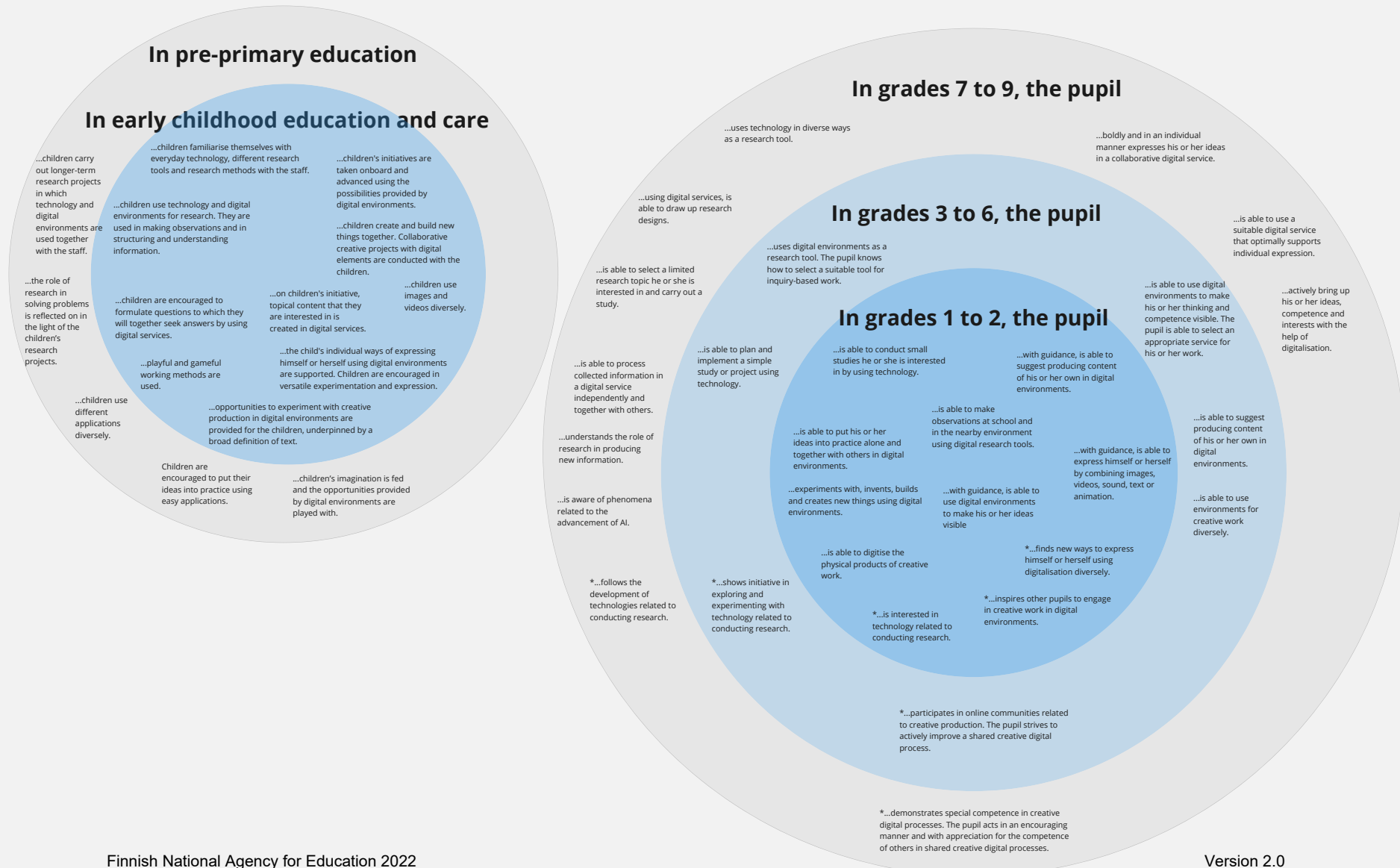
- ...children practise using appropriate search words.
- ...children practise the basic use of a browser.
- ...children familiarise themselves with information searches by finding information on children's interests as well as topical themes and phenomena.
- ...children discuss the reliability of information based on a broad concept of text.
- ...children familiarise themselves with the basic use of a browser and conduct image and sound searches.
- ...children and the staff reflect on what problems could be solved by means of technology and how. Children learn to describe how a problem could be solved.
- ...children practise critical evaluation of information.
- ...attention is paid to comprehensiveness of expression when presenting information. Methods suitable for children are used when presenting information: verbally, visually and physically.
- ...gamefully and playfully, children make classifications and groupings in digital services.
- ...connections between matters are created, verbalised and made visible.
- ...in connection with classifications, children familiarise themselves with concept maps and learn to create them digitally together.

In grades 7 to 9, the pupil

In grades 3 to 6, the pupil

In grades 1 to 2, the pupil

- ...knows how to change the settings of the search engine appropriately. The pupil knows how to do reverse image searches.
- ...harjoittelee etsimään ja hyödyntämään eri lähteistä löytäänsä tietoa.
- ...uses a search engine diversely. The pupil knows how to carry out advanced searches.
- ...knows how to create simple search words.
- ...with guidance, is able to search for information on topical issues and phenomena.
- ...practises explaining in his or her own words the information he or she has searched for.
- ...with guidance, the pupil is able to organise, classify and present information.
- ...is able to search for information on topical issues and phenomena.
- ...uses different sources to produce information.
- ...understands that multiple sources and approaches may change interpretations
- ...is able to obtain, process and present information based on research.
- ...is able to draw up a concept map or other infographics.
- ...with guidance, knows how to structure information in a concept map.
- ...knows how to present information using simple tables.
- ...practises making observations and drawing conclusions based on a simulation.
- ...is able to make observations and draw conclusions based on a simulation.
- ...practises using a device's ease of access functionalities in information management.
- ...uses a device's ease of access functionalities in information management.
- ...is able to edit a document produced together with others.
- ...is able to give and receive feedback on produced content. The pupil is able to edit content based on feedback.
- ...produces and uses written text, images and videos independently and in collaboration.
- ...produces short multimodal texts based on the information searched for.
- *...independently searches for information on issues and phenomena he or she is interested in.
- *...is interested in processing and presenting information. The pupil is able to present information in diverse ways, taking advantage of digital environments.
- ...is able to evaluate the reliability of information. The pupil practises justifying his or her evaluation.
- ...with guidance, is able to use sources in different languages.
- ...understands how online information has been created.
- ...knows how to make observations based on different sources of information in his or her everyday life.
- ...knows how to use digital environments to record and present observations.
- ...learns to use the concepts needed in information management.
- ...with guidance, is able to evaluate the reliability of information.
- ...knows how to use a search engine appropriately. The pupil knows how to use image and sound searches.
- ...knows how to change the settings of the search engine appropriately. The pupil knows how to do reverse image searches.



Interaction: Collaboration and participation

Digital competence

ICT competence

4A

